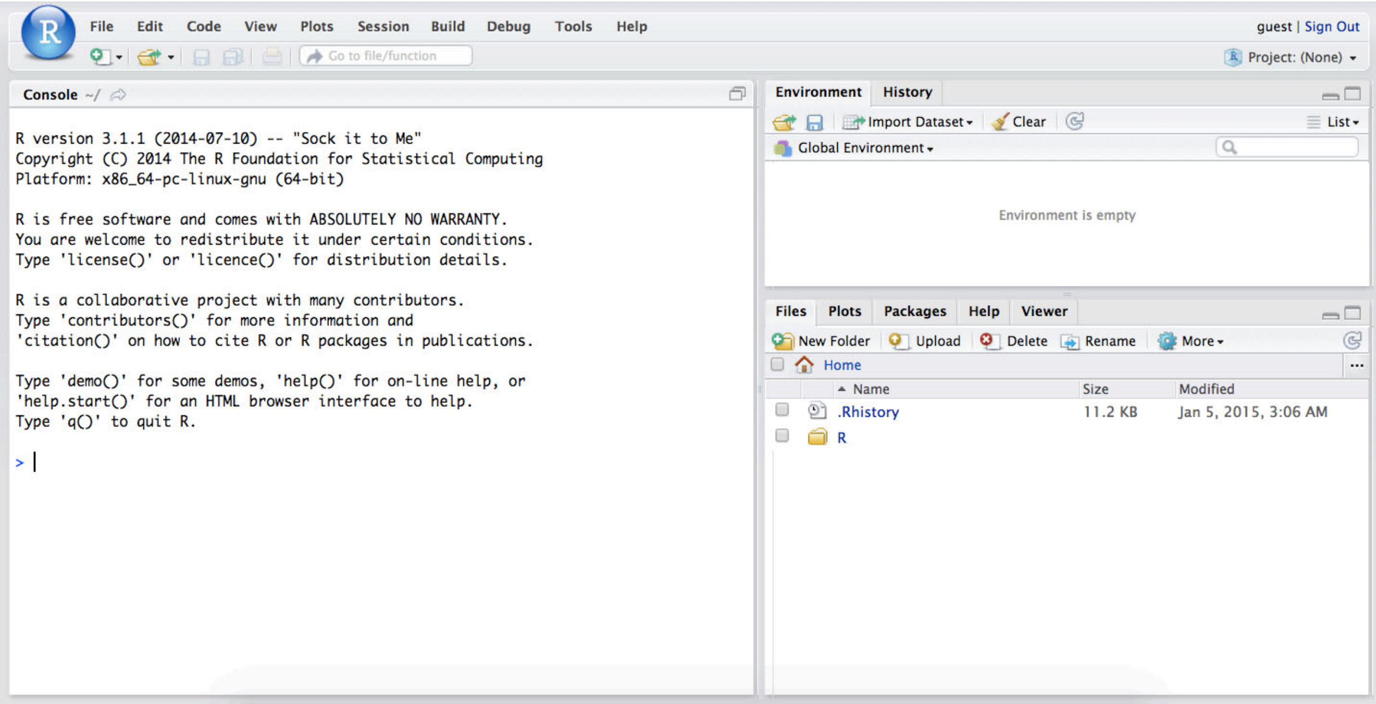
SCUT Data Analysis And Modeling - Lab 1

Introduction to R and RStudio

The goal of this lab is to introduce you to R and RStudio, which you'll be using throughout the course both to learn the statistical concepts discussed in the texbook and also to analyze real data and come to informed conclusions. To straighten out which is which: R is the name of the programming language itself and RStudio is a convenient interface.

As the labs progress, you are encouraged to explore beyond what the labs dictate; a willingness to experiment will make you a much better programmer. Before we get to that stage, however, you need to build some basic fluency in R. Today we begin with the fundamental building blocks of R and RStudio: the interface, reading in data, and basic commands.



The panel in the upper right contains your *workspace* as well as a history of the commands that you've previously entered. Any plots that you generate will show up in the panel in the lower right corner.

The panel on the left is where the action happens. It's called the *console*. Everytime you launch RStudio, it will have the same text at the top of the console telling you the version of R that you're running. Below that information is the *prompt*. As its name suggests, this prompt is really a request, a request for a command. Initially, interacting with R is all about typing commands and interpreting the output. These commands and their syntax have evolved over decades (literally) and now provide what many users feel is a fairly natural way to access data and organize, describe, and invoke statistical computations.

Please note that each of the following practice modules needs to be created in a unified way. Create module at the R prompt (i.e. right after `>` on the console). The template is as follows, you need to manually enter the annotate "Exercise?" before each module, please enter the code in the middle blank. The shortcut key for inserting a module is Ctrl+Alt+I. We propose that you use this function to partition the document in order to increase readability, which will affect your score.

````
###Exercise1
```{r}

```
````

To get you started, enter the following command. You can either type it in manually or copy and paste it from this document.

```
data(pressure)
```

This command instructs R to access the Internal data and fetch some data: changes in temperature and pressure. Click <Promise> of the "values" area in the upper righthand corner of the RStudio window. You should see that the workspace area in the upper righthand corner of the RStudio window now lists a data set called `pressure` that has 19 observations on 2 variables. As you interact with R, you will create a series of objects. Sometimes you load them as we have done here, and sometimes you create them yourself as the byproduct of a computation or some analysis you have performed.

Data and coding exploration

This is a set of numerical records of the temperature and pressure of the same gas. We can take a look at the data by typing its name into the console.

```
pressure
```

What you should see are threer columns of numbers, each row representing a different condition: the first entry in each row is simply the row number (an index we can use to access the data from individual conditions if we want), the second is the temperature, and the third is the pressure at this temperature, respectively. Use the scrollbar on the right side of the console window to examine the complete data set.

Printing out the entire dataset like this is often not a great idea, especially when the sample size is large. Instead we can look at the first few rows using

```
head(pressure)
```

or the last few rows using

```
tail(pressure)
```

Note that the row numbers in the first column are not part of pressure's data. R adds them as part of its printout to help you make visual comparisons. You can think of them as the index that you see on the left side of a spreadsheet. In fact, the comparison to a spreadsheet will generally be helpful. R has stored pressure's data in a kind of spreadsheet or table called a *data frame*.

You can see the dimensions of this data frame by typing:

```
dim(pressure)
```

```
## [1] 19  2
```

This command should output `[1] 19 2`, indicating that there are 19 rows and 2 columns (we'll get to what the `[1]` means in a bit), just as it says next to the object in your workspace. You can see the names of these columns (or variables) by typing:

```
names(pressure)
```

```
## [1] "temperature" "pressure"
```

You should see that the data frame contains the columns `temperature` and `pressure` . At this point, you might notice that many of the commands in R look a lot like functions from math class; that is, invoking R commands means supplying a function with some number of arguments. The `dim` and `names` commands, for example, each took a single argument, the name of a data frame.

Let's start to examine the data a little more closely. We can access the data in a single column of a data frame separately using a command like

```
pressure$temperature
```

This command will only show the temperature value of each condition.

One advantage of RStudio is that it comes with a built-in data viewer. Click on the name `pressure` in the *Environment* pane (upper right window) that lists the objects in your workspace. This will bring up an alternative display of the data set in the *Data Viewer* (upper left window). You can close the data viewer by clicking on the *x* in the upper lefthand corner.

Exercises

Exercise 1 Follow and complete R language basics 1-3 in http://www.cookbook-r.com/Basics/ (http://www.cookbook-r.com/Basics/)

Exercise 2 What command would you use to get some specific data? Please show your code and results

R can select the data range separately, that is, the number of columns and the number of rows. For example, if we want to select the data in the fourth row and second column, we need to type:

```
pressure[4,2]
```

The left represents the number of rows, and the right represents the number of columns. It also allows one side to be empty, representing a whole row or a whole column. For example, if you need to select the second column, you can type it like this:

```
pressure[,2]
```

It is very important to select and distinguish the data reasonably.

Exercise 3 Are temperature and pressure related under different conditions?? How would you describe it? (hint: plot their relationship first)

Notice that the way R has printed these data is different. When we looked at the complete data frame, we saw 19 rows, one on each line of the display. These data are no longer structured in a table with other variables, so they are displayed one right after another. Objects that print out in this way are called *vectors*; they represent a set of numbers. R has added numbers in [brackets] along the left side of the printout to indicate locations within the vector. For example, `300` follows `[1]`, indicating that `300` is the first entry in the vector. And if `[17]` starts a line, then that would mean the first number on that line would represent the 17th entry in the vector.

R has some powerful functions for making graphics. We can create a simple plot of the values of temperature and pressure under different conditions with the command

```
plot(x = pressure$temperature, y = pressure$pressure)
```

By default, R creates a scatterplot with each x,y pair indicated by an open circle. The plot itself should appear under the *Plots* tab of the lower right panel of RStudio. Notice that the command above again looks like a function, this time with two arguments separated by a comma. The first argument in the plot function specifies the variable for the x-axis and the second for the y-axis. If we wanted to connect the data points with lines, we could add a third argument, the letter `l` for **l**ine.

```
plot(x = pressure$temperature, y = pressure$pressure, type = "l")
```

We hope to have a more concise and clear axis identification, so we need to continue to add some argument settings.

```
plot(x = pressure$temperature, y = pressure$pressure, xlab = "T", ylab = "P", type = "l")
```

You might wonder how you are supposed to know that it was possible to add that fifth argument. Thankfully, R documents all of its functions extensively. To read what a function does and learn the arguments that are available to you, just type in a question mark followed by the name of the function that you're interested in. Try the following.

```
?plot
```

Notice that the help file replaces the plot in the lower right panel. You can toggle between plots and help files using the tabs at the top of that panel.

Exercise 4 Now, Make a plot of volume change with temperature. Knowing that the relationship between temperature and pressure is PV=nRT, P is pressure, T is temperature, V is volume, R=8.31, assuming n is 1mol, please calculate the gas volume corresponding to each temperature value, please pay attention to P The unit is mmHg, and the unit of T is Celsius.

Finally, in addition to simple mathematical operators like subtraction and division, you can ask R to make comparisons like greater than, `>`, less than, `<`, and equality, `==`. Here, we've asked R to create *logical* data, data where the values are either `TRUE` or `FALSE`. In general, data analysis will involve many different kinds of data types, and one reason for using R is that it is able to represent and compute with many of them.

Women's height and weight

In this section you're on your own, meaning that there is less handholding with the code. Using what you learned so far you are expected to be able to figure out the necessary code to answer the questions below. Of course, you are not really on your own?, you are welcome and encouraged to ask questions of your TAs or any of the other students in your lab section.

Earlier you recreated some of the displays and preliminary analysis of pressure’s data. Your will now repeat those steps, but for the height and weight of fifteen different women. Load the present day data with the following command:

```
data("women")
```

The data are stored in a data frame called `women` .

Exercise 5 What data does the "height" column have in this data set? What are the dimensions of the data frame? What are the variables?

Exercise 6 Plot the squared height (y axis) vs. weight (x axis) for this data set. What is the overal pattern? Include the plot in your response.

Exercise 7 Which woman (serial number) has the highest BMI index? Does it meet the health value? Tip: BMI = weight (kg) divided by the square of height (m) You can refer to the help files or the R reference card (http://cran.r-project.org/doc/contrib/Short-refcard.pdf (http://cran.r-project.org/doc/contrib/Short-refcard.pdf)) to find helpful functions. Hint: please note that the data set's height unit is inches, and the weight unit is lbs. You will need to convert units.

That was a short introduction to R and RStudio, but we will provide you with more functions and a more complete sense of the language as the course progresses. Feel free to browse around the websites for R (http://www.r-project.org/) and RStudio (http://rstudio.org/) if you’re interested in learning more, or see http://tryr.codeschool.com (http://tryr.codeschool.com/) for a more advanced introduction to R.You can also check health indicators through the website https://www.cancer.org/cancer/cancer-causes/diet-physical-activity/body-weight-and-cancer-risk/adult-bmi.html (https://www.cancer.org/cancer/cancer-causes/diet-physical-activity/body-weight-and-cancer-risk/adult-bmi.html)

Submission

When you are done, regarding the submission of the lab report, please note the following:

1. The report needs to include Exercise1-7;

2. Please enter your student IDs and names of your group at the top of the report;

3. Please submit the lab report within 24 hours after class;

4. The submitted code must be able to run successfully at one time;

5. Please send the lab report to peanutboom.albert@gmail.com in HTML and RMD format, and for each group only one of you send the lab report;

6. Please fill in the name of the email and file in accordance with "student ID 1-name 1-student ID 2-name 2 - lab number - course name". For example, 2021XXXXXXXX-Jack-2021XXXXXXXX -Albert-lab1-Data Analysis and Modeling.

This text refers to some content from Duke University, for which we are grateful.